From Ballots to Coalitions

A Data-Driven Analysis of Austrian Parliamentary Elections

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1 Executive Summary

Austrian parliamentary elections determine which parties enter the *Nationalrat*, Austria's 183-seat lower house, and which coalition governments become mathematically possible. Two recent elections are analysed here: the **2019 snap election**, called after the Ibiza affair brought down the ÖVP–FPÖ government, and the **2024 election**, in which the Freedom Party (FPÖ) finished first for the first time in its history.

This report walks through a complete data pipeline built in R: automated reading and cleaning of the official result files, consistency checks across all geographic levels, regional performance analysis, seat allocation with the D'Hondt method, coalition arithmetic and a Monte Carlo simulation of polling uncertainty.

Key Findings at a Glance:

- **The 2024 election marked a historic shift.** The FPÖ won 28.8% of the valid votes and 57 of 183 seats, its best result ever and the first time it finished first in a national election.

- **Winning the most votes is not the same as governing.** With a majority threshold of 92 seats, three two-party majorities existed in 2024: FPÖ–ÖVP (108), FPÖ–SPÖ (98), ÖVP–SPÖ (92). An FPÖ–ÖVP majority was therefore possible, and the two parties did negotiate in early 2025, but the talks failed. The government formed in March 2025 is a three-party coalition of ÖVP, SPÖ and NEOS without the largest party.

- **Regional differences are large.** At district level the FPÖ ranged from 9.45% in Wien 7.,Neubau to 44.81% in Feldkirchen, a gap of 35.4 percentage points.

- **Small polls are unreliable for close races.** In polls of 200 respondents, the ÖVP is level with or ahead of the FPÖ in 33.0% of simulated polls, although the true gap was 2.6 percentage points.

About this Report

Data source	Official Austrian election results (Federal Ministry of the Interior)
Elections	Nationalratswahl 2019 and 2024, Bundespräsidentenwahl 2022
Methods	Data pipeline, consistency checks, D'Hondt, Monte Carlo simulation
Tools	R (`openxlsx`, `ggplot2`, `kableExtra`)
Author	Filip Popovic

2 The Data Pipeline

2.1 Source & Structure

Austrian election results are published by the Federal Ministry of the Interior as Excel files with one row per geographic unit: the national total, the nine federal states (*Bundesländer*), the regional constituencies (*Wahlkreise*, parliamentary elections only), the districts (*Bezirke*) and the municipalities (*Gemeinden*). Postal votes (*Wahlkarten*) appear as separate rows at the district, constituency or state level. Each row carries a geographic code (GKZ) whose structure reveals the level: for example `G00000` is Austria, `G30000` Lower Austria, `G31000` the district of Hollabrunn and `G31001` one of its municipalities. The columns contain eligible voters, ballots cast, invalid and valid votes, and the votes and vote shares of every party.

The analysis covers three elections:

- **Nationalratswahl 2019**: snap parliamentary election after the Ibiza affair
- **Nationalratswahl 2024**: regular parliamentary election
- **Bundespräsidentenwahl 2022**: presidential election, available in three versions (an erroneous first publication, a corrected but still faulty version, and a fixed version reconstructed by the course instructor). It serves as a test case for the validation step.

2.2 Automated Ingestion & Cleaning

The pipeline finds all `.xlsx` files in the data folder with `list.files()`, reads them into a named list and cleans them in a single pass:

- `read.clean()` removes header and footer rows that have no geographic code.
- `set.colnames()` gives the first six columns a common name across all files and builds the party columns from the Excel header: the vote column keeps the party name and the following percentage column becomes `"<party> %"`.
- `fix.spaltenmodes()` converts every column that contains only numbers to numeric.
- `list.cleaner()` sets a party's vote share to `NA` wherever its vote count is missing, i.e. wherever the party did not run.
- `gkz.ebene()` derives the geographic level of each row from its GKZ. All later steps use this function instead of separate regular expressions.

As a check, the cleaned data frames were compared with the reference objects provided in the course: names, dimensions, codes and all vote counts are identical.

3 Consistency Validation

Before any analysis can be trusted, the data must be verified. The function `checke.wahlergebnis()` runs six checks and returns one row per failing unit, together with the columns that do not add up.

3.1 The Six Checks

Check 1. Ballots cast = valid + invalid votes in every row.

Check 2. Valid votes = sum of all party votes in every row.

Check 3. Vote shares are correct: each published share must match $100 \cdot$ votes/valid votes up to 0.01 percentage points.

Check 4. Austria = sum of the nine federal states for all count columns.

Check 5. Federal state = sum of its districts. In parliamentary elections, postal votes that cannot be assigned to a district are counted at state or constituency level and are added here. In the presidential election the state-level postal vote row is only a subtotal of the district postal votes and is therefore not added a second time.

Check 6. District = sum of its municipalities, including the district's postal votes.

3.2 Validation Results

Table 1: Number of failing geographic units per check and dataset.

Dataset	Check 1	Check 2	Check 3	Check 4	Check 5	Check 6	Total
Presidential election 2022 (error)	0	0	0	0	1	3	4
Presidential election 2022 (corrected)	0	2	0	0	1	1	4
Presidential election 2022 (fixed)	0	0	0	0	0	0	0
Parliamentary election 2019	0	0	0	0	0	0	0
Parliamentary election 2024	0	0	0	0	0	0	0

Table 2: Inconsistencies found in the two faulty versions of the 2022 presidential election results.

Dataset	Check	GKZ	Gebiet	Inconsistent columns
(error)	Check 5	G20000	Kärnten	Abgegebene, Ungueltige
(error)	Check 6	G20200	Villach Stadt	Abgegebene, Ungueltige, Gueltige, Dr. Alexander Van der Bellen
(error)	Check 6	G20500	Sankt Veit an der Glan	Abgegebene, Ungueltige
(error)	Check 6	G50100	Salzburg Stadt	Wahlberechtigte
(corrected)	Check 2	G20200	Villach Stadt	all party columns
(corrected)	Check 2	G20201	Villach	all party columns
(corrected)	Check 5	G20000	Kärnten	Dr. Alexander Van der Bellen
(corrected)	Check 6	G50100	Salzburg Stadt	Wahlberechtigte

The checks behave as expected. The first publication of the 2022 presidential election contains 4 inconsistent units, mainly districts whose totals do not match their municipalities (Villach, Sankt Veit an der Glan, Salzburg). The "corrected" version fixes some of these but introduces new errors, for example in Villach, where the valid votes no longer equal the sum of the candidates' votes. Only the fixed version passes all checks, as do both parliamentary elections.

One detail is worth noting. In the 2024 file the vote shares are cut off after two decimals instead of being rounded: for example the ÖVP is listed with 26.26% although its exact share is 26.27%. Of the 12,343

share values that differ from correct rounding, 12,343 are exactly the truncated value. A check that requires the shares to add up to exactly 100 would therefore flag almost every row. Check 3 instead compares each share with its exact value and allows a deviation of 0.01 percentage points, which covers both rounding and truncation.

4 Regional Performance Analysis

The function `ana.wahl()` finds, for each party, the geographic unit with its best and worst vote share at a chosen level (federal state, district or municipality). Units where a party did not run are ignored. The ranking uses the exact share (votes divided by valid votes) rather than the rounded value from the file, so ties are rare. Remaining ties are broken in favour of the unit with more valid votes, i.e. the larger and more representative one.

4.1 2024 Election: Federal State Level

Table 3: Best and worst federal state for each party, Nationalratswahl 2024 (vote shares in %).

Partei	Top Region	Top %	Flop Region	Flop %
ÖVP	Salzburg	31.55	Wien	17.42
SPÖ	Wien	29.88	Vorarlberg	13.09
FPÖ	Kärnten	38.38	Wien	20.66
GRÜNE	Wien	12.34	Burgenland	4.70
NEOS	Vorarlberg	12.64	Burgenland	6.47
BIER	Vorarlberg	2.38	Salzburg	1.53
MFG	Oberösterreich	0.84	Wien	0.20
BGE	Burgenland	0.08	Burgenland	0.08
LMP	Vorarlberg	0.75	Burgenland	0.45
GAZA	Wien	1.12	Steiermark	0.15
KPÖ	Wien	3.83	Burgenland	1.28
KEINE	Kärnten	0.70	Wien	0.44

4.2 2024 Election: District Level

Table 4: Best and worst district for each party, Nationalratswahl 2024 (vote shares in %).

Partei	Top Region	Top %	Flop Region	Flop %
ÖVP	Zwettl	40.94	Wien 15.,Rudolfsheim-Fünfhaus	11.61
SPÖ	Wien 15.,Rudolfsheim-Fünfhaus	34.88	Lienz	10.59
FPÖ	Feldkirchen	44.81	Wien 7.,Neubau	9.45
GRÜNE	Wien 7.,Neubau	22.30	Wolfsberg	3.12
NEOS	Wien 1.,Innere Stadt	19.00	Güssing	4.87
BIER	Wien 11.,Simmering	3.78	Wien 1.,Innere Stadt	0.95
MFG	Rohrbach	1.18	Wien 15.,Rudolfsheim-Fünfhaus	0.14
BGE	Oberpullendorf	0.37	Eisenstadt - Umgebung	0.03
LMP	Rust Stadt	0.94	Sankt Johann im Pongau	0.29
GAZA	Wien 10.,Favoriten	2.40	Rust Stadt	0.00
KPÖ	Wien 15.,Rudolfsheim-Fünfhaus	6.54	Hermagor	0.87
KEINE	Perg	0.92	Wien 1.,Innere Stadt	0.19

Regional differences grow with the level of detail. Between federal states, the gap between a major party's best and worst result is at most 17.7 percentage points (FPÖ). Between districts it reaches 35.4 points for the FPÖ and 29.3 points for the ÖVP. The clearest divide runs between Vienna's inner districts, where the Greens and NEOS are strongest, and rural districts in Carinthia and Lower Austria, where the FPÖ and ÖVP dominate.

4.3 2019 vs. 2024: A Shift in the Political Map

Table 5: Best and worst federal state for each party, Nationalratswahl 2019 (vote shares in %). Parties that ran in only one state have the same top and flop region.

Partei	Top Region	Top %	Flop Region	Flop %
ÖVP	Salzburg	46.38	Wien	24.63
SPÖ	Burgenland	29.38	Tirol	13.02
FPÖ	Kärnten	19.78	Wien	12.83
NEOS	Vorarlberg	13.56	Burgenland	4.88
JETZT	Wien	2.96	Burgenland	1.26
GRÜNE	Wien	20.69	Burgenland	8.08
KPÖ	Steiermark	1.25	Burgenland	0.40
WANDL	Vorarlberg	0.78	Burgenland	0.26
BZÖ	Kärnten	0.24	Kärnten	0.24
BIER	Wien	0.60	Wien	0.60
CPÖ	Burgenland	0.14	Burgenland	0.14
GILT	Vorarlberg	0.45	Tirol	0.24
SLP	Oberösterreich	0.04	Oberösterreich	0.04

Comparing the two elections at state level shows how much the political map changed. The FPÖ's best state result roughly doubled, from 19.78% to 38.38% (both in Carinthia), and even its weakest state, Vienna, rose from 12.83% to 20.66%. The ÖVP shows the mirror image: its best result in Salzburg fell from 46.38% to 31.55%. The Greens, who returned to parliament in 2019 after missing the 4% threshold in 2017, remained strongest in Vienna but lost ground there as well (20.69% to 12.34%).

5 Seat Allocation via the D'Hondt Method

5.1 How the D'Hondt Method Works

The actual allocation of the 183 seats in the Nationalrat uses three stages (regional constituencies, federal states and a nationwide stage); the D'Hondt method is only used in the last one. Following the course assignment, this report uses a simplified version that allocates all seats with D'Hondt at the national level. For 2019 and 2024 this simplification gives exactly the official seat distribution.

1. Parties with less than **4%** of the valid votes are excluded. (In reality a party can also qualify by winning a seat in a regional constituency; this did not occur in 2019 or 2024.)
2. Each remaining party's votes are divided by $1, 2, 3, \ldots, 183$.
3. The 183 largest quotients receive one seat each.
4. If several quotients tie at the last seat(s), the remaining seats are drawn by lot among the tied parties and a warning is issued.

The implementation computes the full matrix of quotients with a single `outer()` call. The smallest successful quotient is the 183rd largest value; every quotient above it is a seat, and only the quotients exactly at this value can lead to a tie. The function was tested with small examples, including ties for the last seat, in addition to the two real elections.

5.2 Seat Distribution: 2024

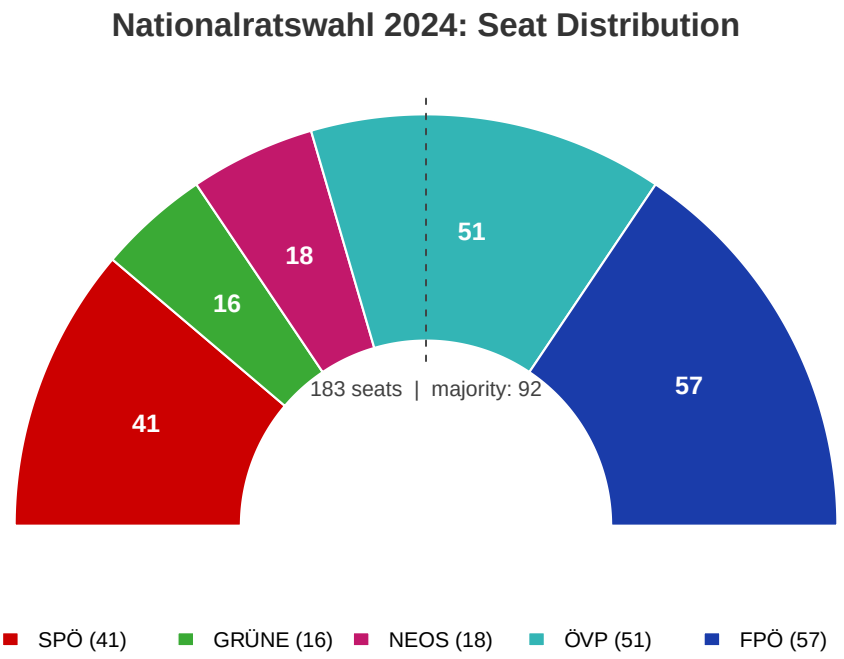


Figure 1: Seat distribution after the Nationalratswahl 2024, with parties arranged from left to right as in parliament. The dashed line marks the middle of the hemicycle: a coalition needs at least 92 of the 183 seats.

Table 6: Vote shares and seats, Nationalratswahl 2024. Because parties below 4% receive no seats, the parties in parliament hold a slightly larger share of seats than of votes.

Party	Votes	Seats	Seat share
FPÖ	28.8%	57	31.1%
ÖVP	26.3%	51	27.9%
SPÖ	21.1%	41	22.4%
NEOS	9.1%	18	9.8%
GRÜNE	8.2%	16	8.7%

5.3 Seat Distribution: 2019

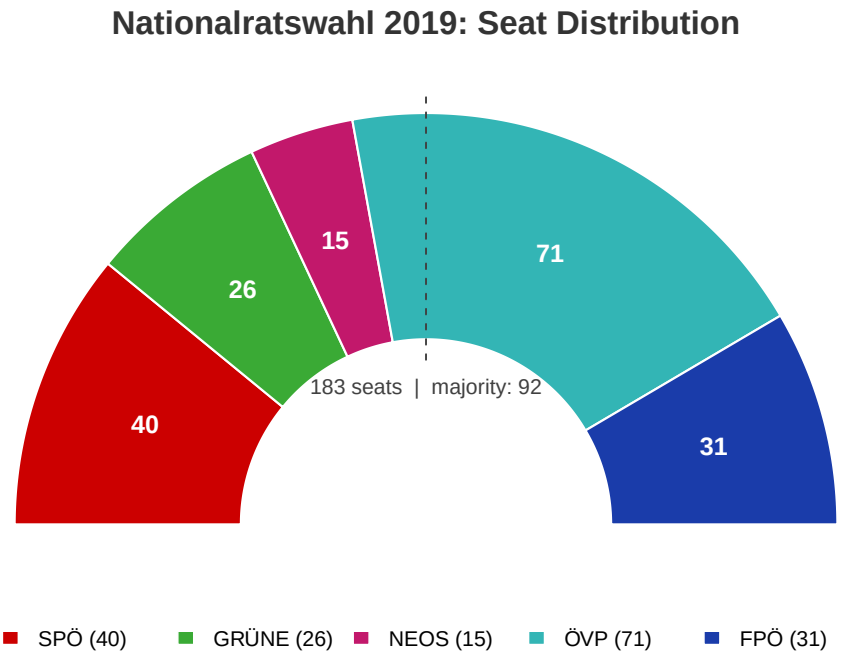


Figure 2: Seat distribution after the Nationalratswahl 2019. The ÖVP was the clear largest party and later formed the first ÖVP–Greens coalition at federal level.

Table 7: Vote shares and seats, Nationalratswahl 2019.

Party	Votes	Seats	Seat share
ÖVP	37.5%	71	38.8%
SPÖ	21.2%	40	21.9%
FPÖ	16.2%	31	16.9%
GRÜNE	13.9%	26	14.2%
NEOS	8.1%	15	8.2%

6 Coalition Modelling

6.1 Mathematical Feasibility

A government needs an **absolute majority** of $\lfloor 183/2 \rfloor + 1 = 92$ seats. The function `finde.koalitionen()` lists all combinations of two up to `nmax` parties with `combn()` and keeps those with a majority. `plot.koalitionsvarianten()` shows them as stacked bars, sorted by total seats, with the largest party of each coalition at the bottom of its bar and the total above it. The charts below show all majorities with up to three parties; bars with two segments are the two-party majorities. Whether a coalition is politically realistic is not part of this calculation.

6.2 Coalitions: 2024

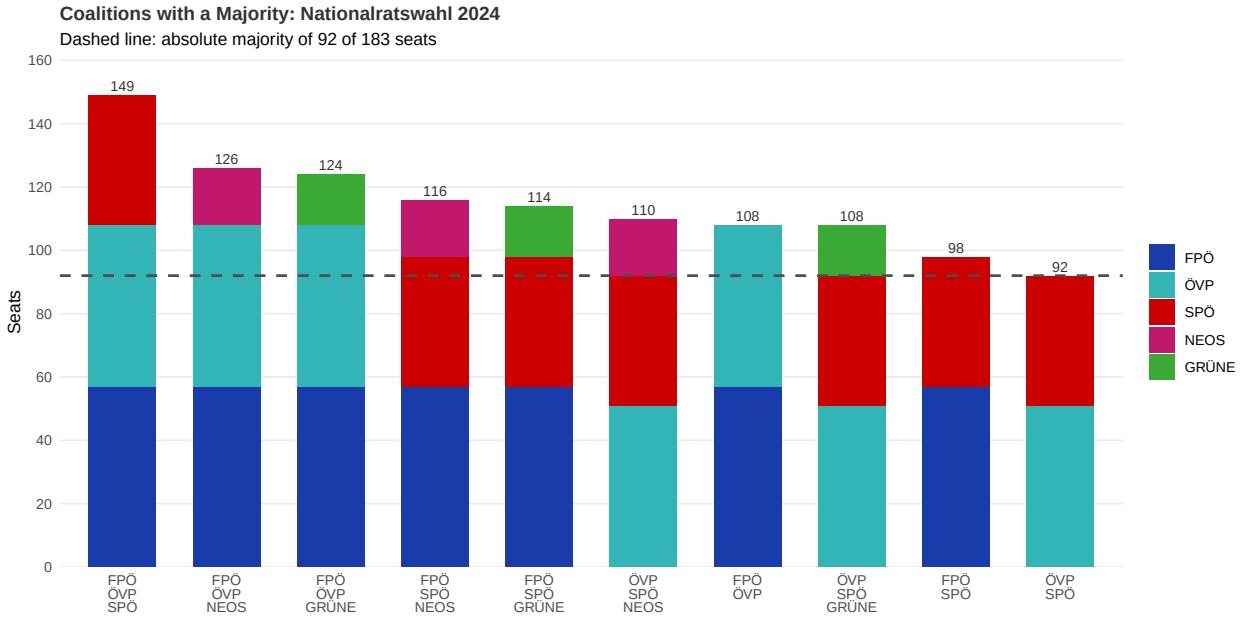


Figure 3: All coalitions of two or three parties with an absolute majority after the 2024 election.

In 2024 no party came close to a majority on its own. Three two-party majorities existed: FPÖ–ÖVP (108), FPÖ–SPÖ (98), ÖVP–SPÖ (92). The FPÖ was part of two of them, so arithmetically it was not isolated. The outcome was decided by politics: after talks between ÖVP, SPÖ and NEOS broke down in January 2025, the FPÖ led coalition negotiations with the ÖVP, which failed in February 2025. The government formed in March 2025 is a coalition of ÖVP, SPÖ and NEOS with 110 seats. ÖVP and SPÖ alone would have had exactly 92 seats, the smallest possible majority.

6.3 Coalitions: 2019

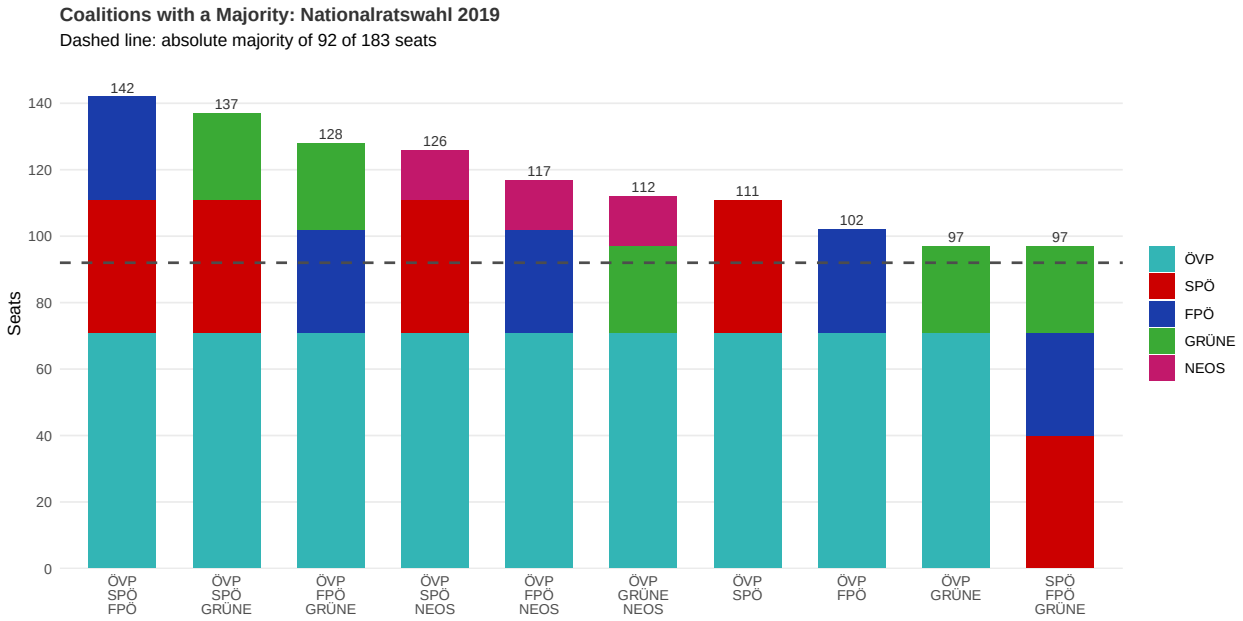


Figure 4: All coalitions of two or three parties with an absolute majority after the 2019 election.

In 2019 the ÖVP was part of every two-party majority: ÖVP–SPÖ (111), ÖVP–FPÖ (102), ÖVP–GRÜNE (97). It could therefore choose its partner. It chose the Greens, and with 97 seats this two-party coalition had a clear majority.

7 Polling Simulation

7.1 The Uncertainty of Small Samples

How much can a single opinion poll tell us? The function `simuliere.umfrage()` answers this by simulation. It draws $n = 100{,}000$ polls of a given size from a multinomial distribution whose probabilities are the true 2024 vote shares, and returns the simulated shares with the parties ordered by their median result.

Formally, for a poll of size m with true party proportions $\mathbf{p} = (p_1, \dots, p_k)$:

$$(X_1, \dots, X_k) \sim \text{Multinomial}(m, \mathbf{p}), \qquad \hat{p}_j = X_j / m.$$

The main simulation uses 200 respondents per poll, as specified in the course assignment. Published polls in Austria usually interview between 800 and 2,000 people, so a second simulation with 1,000 respondents is used for comparison.

7.2 Simulated Poll Distribution: 2024 Proportions

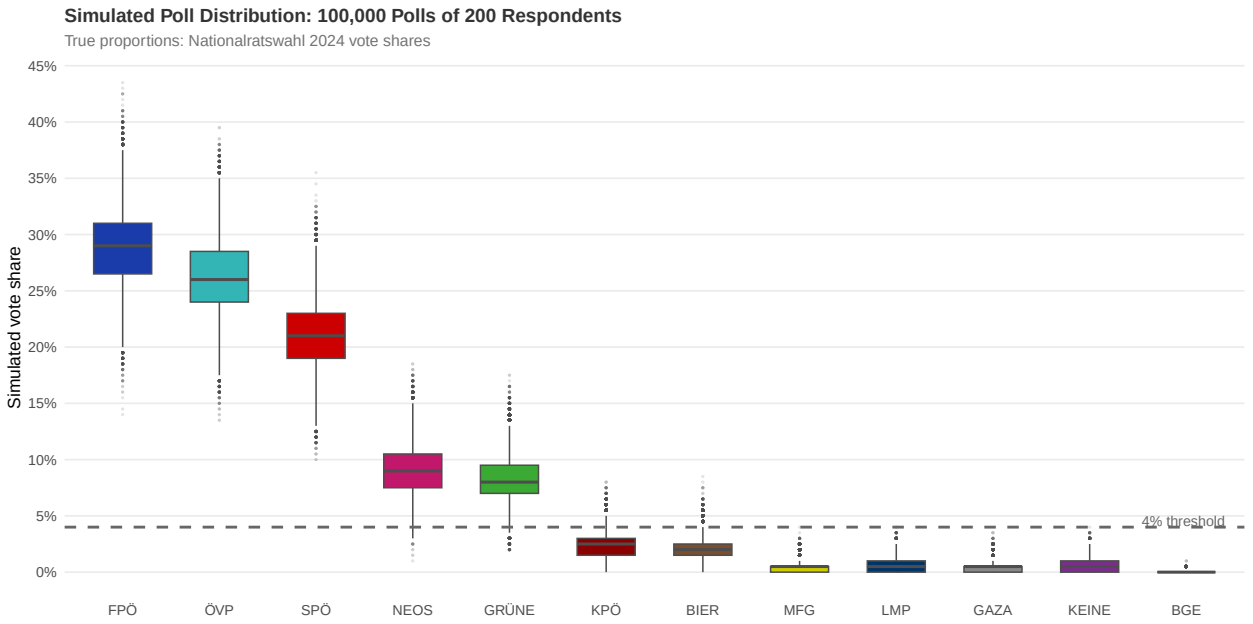


Figure 5: Distribution of 100,000 simulated polls with 200 respondents each, based on the 2024 result. Boxes show the interquartile range, whiskers extend to 1.5 times the IQR, and the dashed line marks the 4% threshold.

Table 8: Summary of 100,000 simulated polls with 200 respondents. The middle 90% column gives the range in which 90% of the polls fall; P(below 4%) is the share of polls in which the party appears below the parliamentary threshold.

Party	True share	Median	Middle 90%	P(below 4%)
FPÖ	28.8%	29.0%	23.5% to 34.0%	0.0%
ÖVP	26.3%	26.0%	21.0% to 31.5%	0.0%
SPÖ	21.1%	21.0%	16.5% to 26.0%	0.0%
NEOS	9.1%	9.0%	6.0% to 12.5%	0.1%

GRÜNE	8.2%	8.0%	5.0% to 11.5%	0.6%
KPÖ	2.4%	2.5%	1.0% to 4.5%	89.1%
BIER	2.0%	2.0%	0.5% to 4.0%	94.9%
MFG	0.4%	0.5%	0.0% to 1.0%	100.0%
LMP	0.6%	0.5%	0.0% to 1.5%	100.0%
GAZA	0.4%	0.5%	0.0% to 1.0%	100.0%
KEINE	0.6%	0.5%	0.0% to 1.5%	100.0%
BGE	0.0%	0.0%	0.0% to 0.0%	100.0%

With 200 respondents a single poll is a rough guide at best. The FPÖ led the ÖVP by 2.6 percentage points, yet the ÖVP is level or ahead in 33.0% of the simulated polls, so even the ranking of the two largest parties is uncertain. With 1,000 respondents this drops to 14.0%.

For the 4% threshold the picture is clearer, because no party was close to it in 2024: NEOS and the Greens (9.1% and 8.2%) appear below 4% in only 0.1% and 0.6% of the polls, while the KPÖ (2.4%) appears above it in 10.9%. With 1,000 respondents these error rates fall to 0.00% for the Greens and 0.1% for the KPÖ. A party with a true share close to 4%, however, would appear on the wrong side of the threshold in a large share of polls at any realistic sample size.

8 Conclusion

This project builds a complete electoral data pipeline in R, from the raw official result files to seat allocation, coalition arithmetic and a polling simulation, without specialised election packages.

The validation step shows why such checks matter: both faulty versions of the 2022 presidential results contain inconsistencies that are easy to miss by eye, while the fixed version and both parliamentary elections are internally consistent.

The 2024 election illustrates that winning the most votes does not guarantee a place in government. The FPÖ finished first with 57 seats and had a possible majority with the ÖVP, but after failed negotiations the government was formed by ÖVP, SPÖ and NEOS. Seat arithmetic defines which coalitions are possible; which of them is formed is a political decision.

The polling simulation adds a further point: even if the true vote shares were known exactly, a single small poll could easily get the order of close parties wrong. Polls should therefore be read as ranges, not as point predictions.

All functions work on any of the five files and adapt to the number of parties and the structure of the geographic codes. A future election can be added by placing its result file in the data folder, provided the ministry keeps the same file format.

This analysis was created by Filip Popovic as part of a data analyst portfolio. For questions or inquiries, ***contact me at fpopovic01@gmail.com****.*